IT2011-AI & ML

Lab sheet -01

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Part 1: Traditional Programming vs AI/ML

Task 1A – Rule-Based Programming

In [1]:

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print(f"{num} is even.")
else:
    print(f"{num} is odd.")
```

```
Enter a number: 96
96 is even.
```

Limitations: -

1.A programmer explicitly writes rules to process data and cannot adapt new patterns

2.Only works for the specific task of even/odd classifications

3.Cannot learn from data or improve its performance

Task 1B – ML-Based Classification

In [22]:

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier


data = {
    "value": [1, 2, 3, 4, 5, 6, 7, 8],
    "label": [1, 0, 1, 0, 1, 0, 1, 0]  # 1 = odd, 0 = even
}
df = pd.DataFrame(data)


x = df[["value"]]
y = df["label"]
model = DecisionTreeClassifier()
model.fit(x, y)


test_values = pd.DataFrame({"value": [10, 11, 13]})


print("Predicted labels (1 = odd, 0 = even):")


for num in test_values["value"]:
    if num % 2 == 0:
        print(f"{num} is even(0).")
    else:
        print(f"{num} is odd(1).")
```

```
Predicted labels (1 = odd, 0 = even):
10 is even(0).
11 is odd(1).
13 is odd(1).
```

```
In [24]:  import pandas as pd
          from sklearn.tree import DecisionTreeClassifier


          data = {
              "value": list(range(1,101)),
              "label": [x%2 for x in range(1,101)]
          }
          df = pd.DataFrame(data)


          x = df[["value"]]
          y = df["label"]
          model = DecisionTreeClassifier()
          model.fit(x, y)


          test_values = pd.DataFrame({"value": [10, 11, 13]})
          print("Predicted labels (1 = odd, 0 = even):")
          print(model.predict(test_values))

          Predicted labels (1 = odd, 0 = even):
          [0 1 1]
```

difference from rule-based logic

1.The ML approach learns patterns from data rather than following explicit rules

2.Can handle patterns, noise and real-world data

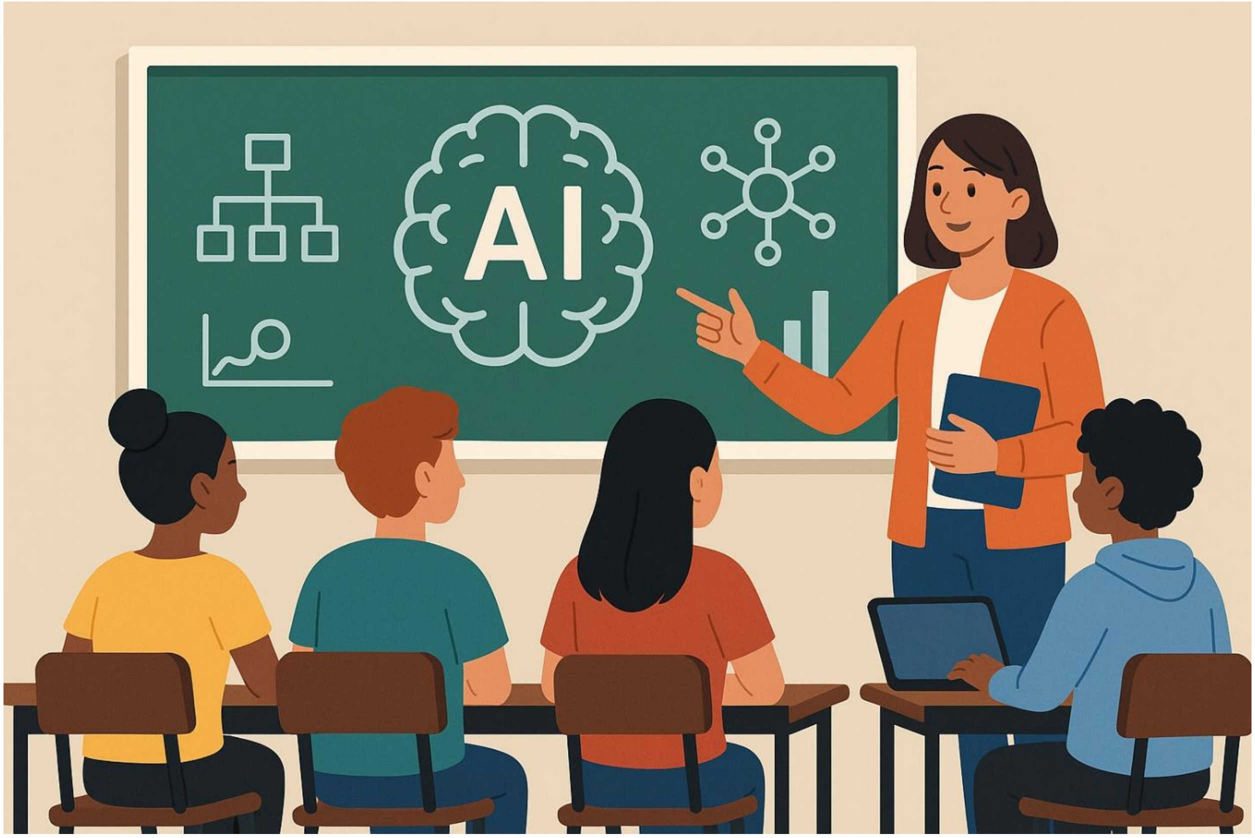
3.More flexible and scalable than hardcoded logic

Part 2: Generative AI and Prompt Engineering

1)- A short story or poem

In halls of knowledge, circuits spark,
AI learns to leave its mark.
From code and data, insights grow,
Guiding minds with what they know,
A future bright begins to glow

- An AI-generated image



2)- Write a short 5-line poem about Artificial Intelligence in schools

In classrooms bright, AI takes its place,
A guiding hand in learning's race.
It tailors paths for every mind,
With knowledge vast and answers kind.
A future shaped by code and grace

- Describe a futuristic classroom in 2035 using vivid imagery

walls made of transparent smart glass that shift scenes from lush rainforests to distant galaxies, immersing students in their lessons. Desks are sleek, adaptive pods that adjust to each student's posture, with holographic screens floating above, projecting interactive 3D models. AI assistants, embodied as gentle glowing orbs, drift quietly, offering personalized help without interrupting. The air hums softly with quiet drones delivering materials and snacks on demand. Natural light blends seamlessly with ambient neon hues, creating a calm yet electrifying atmosphere where learning feels boundless and alive.

-"What are the ethical concerns of AI in Education?"

- **Privacy:** AI systems often collect and analyze large amounts of student data, raising worries about how that data is stored, who can access it, and whether students' privacy is truly protected.

- **Bias and Fairness:** If AI algorithms are trained on biased data, they can unintentionally reinforce inequalities, giving some students unfair advantages or disadvantages.

- **Transparency:** Many AI systems are complex "black boxes," making it hard for educators and students to understand how decisions or recommendations are made.

- **Dependency:** Overreliance on AI might reduce human interaction and critical thinking, potentially weakening essential social and cognitive skills.

- **Accessibility and Equity:** Not all schools have equal access to advanced AI tools, which could widen the educational gap between different communities.

3)

Prompt: Explain artificial intelligence.
Response:
Artificial intelligence (AI) is a branch of computer science that creates machines or software that can think, learn, and solve problems like humans do. It uses data and algorithms to recognize patterns, make decisions, and improve over time without being explicitly programmed for every task.

Prompt: Explain AI to a 7-year-old using cartoon characters.
Response:
Imagine your favorite cartoon robot friend, like Wall-E or Baymax. AI is like their brain—it helps them learn new things and help people. Just like how you learn to play games or solve puzzles, AI is a smart computer that learns too, so it can help us with fun stuff like games, stories, or even answering questions!